

Project: Real-Time Collaborative Kanban Board

Build a collaborative task board similar to a simplified version of Trello or Jira where **multiple users can edit tasks in real time**.

1□ Core Requirements

Users should be able to:

- Register
- Login
- Create boards
- Invite members to boards
- Create lists (columns)
- Create tasks
- Move tasks between lists

Example lists:

To Do

In Progress

Done

Authentication can use:

- JSON Web Token

2□ Board Structure

Example schema:

Board

- id
- name
- owner_id
- created_at

BoardMembers

- id
- board_id
- user_id
- role (admin/member)

3▢ Lists (Columns)

Each board contains multiple lists.

Example:

List

- id
- board_id
- title
- position

Example:

- 1 → Todo
- 2 → In Progress
- 3 → Done

4▢ Tasks (Cards)

Users can create tasks within lists.

Task

- id
- list_id
- title
- description
- assigned_to
- due_date
- position
- created_at

Features:

- Create task
- Edit task
- Delete task
- Assign users
- Move task between lists

5▢ Real-Time Updates

When a user:

- moves a task
- edits a task
- creates a task

Other users viewing the board should see **updates instantly**.

Candidates can implement this using:

- Socket.IO
- WebSockets

6 Drag & Drop UI

Frontend should allow:

- drag tasks between lists
- reorder tasks

Libraries that can help:

- React
- react-beautiful-dnd

7 API Design

Example endpoints:

POST /auth/register
POST /auth/login

POST /boards
GET /boards/:id

POST /lists
PATCH /lists/:id

POST /tasks
PATCH /tasks/:id
DELETE /tasks/:id